

Chapter 3

Optically Stimulated Electron Emission: A Powerful Tool for Surface Cleanliness Monitoring

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1 INTRODUCTION

Optically Stimulated Electron Emission (OSEE), also known as photoemission or photoelectric effect, is a phenomenon in which charged particles are released from a material when it absorbs radiant energy. The photoelectric effect commonly is thought of as the ejection of electrons from the surface of a metal

plate when light falls on it. In the broad sense, however, the phenomenon can take place when the radiant energy is in the region of visible or ultraviolet (UV) light, X-ray, or gamma rays; when the material is solid, liquid, or gas; and when the particles released are electrons or ions (charged atoms or molecules).

The photoelectric effect was discovered in 1887 by German physicist, Heinrich Rudolf Hertz [1,2], who observed that UV light changes the lowest voltage at which sparking takes place between given metal electrodes. At the close of nineteenth century, it was established that a cathode ray (produced by an electric discharge in a rarefied-gas atmosphere) consists of discrete particles, called electrons, each bearing an elementary negative charge. In 1900, Philipp Edward Anton Lenard, a German physicist, studying the electrical charges liberated from metal surface when it was illuminated, concluded that these charges were identical to the electrons observed in cathode rays. It was further discovered that the current (given the name photoelectric because it was caused by light rays), made up of electrons released from the metal, is proportional to the intensity of the light causing it for any fixed wavelength of light that is used. In 1902, it was shown that the maximum kinetic energy of an electron in the photoelectric effect is independent of the intensity of the light ray and depends on the frequency of the incident light. The frequency of the incident light determines the energy imparted by the incident light when it strikes any material.

The observations that (1) the number of electrons released in the photoelectric effect is proportional to the intensity of the light and that (2) the frequency, or wavelength, of light determines the maximum kinetic energy of the electrons, indicated a kind of interaction between light and matter that could not be explained in terms of classical physics. The search for an explanation led in 1905 to Albert Einstein's fundamental theory that light, long thought to be wavelike, can be regarded alternatively as composed of discrete particles (now called photons), equivalent to energy quanta.

In 1905, Albert Einstein published a paper that explained experimental data from the photoelectric effect as being the result of light energy being carried in discrete quantized packets. This discovery led to the quantum revolution. Albert Einstein was awarded the Nobel Prize [3] in 1921 for "his discovery of the law of the photoelectric effect." In explaining the photoelectric effect, Einstein assumed that a photon could penetrate matter, where it would collide with an atom. Since all atoms have electrons, an electron would be ejected with great velocity from the atom by the energy of the photon. Originally, most investigations of photoemission centered on the emissive properties of solid metals that are good conductors of electricity. Gradually, with the recognition of the effects of volume and surface impurities, there evolved an interest in the photoelectric emission from insulators, which are nonconductors, and from semiconductors, which conduct electricity only under certain circumstances. The energy distribution of electrons in insulators and semiconductors is different from that of

metals. As a consequence, the photoelectric yield (number of electrons emitted for each incident quantum of radiation) versus energy of the incident photon can be different for materials having the same work function.

In the earlier days of discovery of this phenomenon, it was observed to work only in vacuum. From the time of discovery until the late ~1970s, this technique was a laboratory technique. It was in the late 1970s that the National Aeronautics and Space Administration (NASA) funded research to study whether this technique could be useful for measuring the cleanliness of the outer surface of the Space Shuttle prior to bonding of tiles to the outer surface that were required to protect the Shuttle surface from overheating during reentry into the Earth's atmosphere. Since there was a tremendous advancement in electronics between the 1920s and the late 1970s, this phenomenon was demonstrated to be able to work in ambient environment. The first commercially available systems based on this principle were introduced to the market in 1985 by Photo Acoustic Tech., Inc, a predecessor of Photo Emission Tech., Inc. This principle has been applied very usefully to study the chemical state of solids, specifically to measure/monitor surface cleanliness (thin film contamination), absence/presence of films and measuring the thickness of very thin films (typically less than 15 nm). The following discussion is related to the use of this technique for the aforementioned applications.

In summary, UV light in the short wavelength (~185 nm) range can impart sufficient energy to the electrons in a material thus releasing them from the surface of a solid. This is the OSEE process (or photoelectric emission or photoemission); a material that exhibits this phenomenon is said to be photoemitting, and the ejected electrons are called photoelectrons. For any material to emit electrons, the work function of that material has to be less than the energy imparted by the incident light. Otherwise, there will be no emission from a given material. A simulation of OSEE can be seen at the University of Colorado, Boulder website [4].

2 OSEE PRINCIPLE

The electrons in any material have a certain amount of force that binds them to the atom. If the surface of any material, particularly a metal surface, is exposed to light, the electrons absorb the energy from the light.

Light is made up of photons with a certain amount of energy. The energy of light is given by $E = hf$, where h is Planck's constant (the value of Planck's constant is typically 4-5 eV) and f is the frequency of light. Higher intensity of light means more photons. The energy of the photon of light is used to overcome the binding energy of an electron.

This binding energy is known as the work function. Work function (ϕ) can be described as the minimum energy required to release an electron. If the energy of the photon is just sufficient to overcome the work function of a given material, the electron can be released but it will have no kinetic energy.

Thus, $\phi = hf_0$, where f_0 is the threshold frequency of light that will provide sufficient energy to release an electron from a given material.

When the frequency of incident light is $<f_0$, the energy gained by an electron is not sufficient to release the electron. Hence, no photoelectrons are released below the threshold frequency.

When the frequency of incident light is $>f_0$, the energy gained by an electron is enough to release the electron. Hence, some photoelectrons are released. The law of conservation of energy dictates that all the energy of the photon will be transferred onto the electron. Some of the energy is used to enable the electron to escape from its orbit in the atom. The remainder of the energy from the photon of light is converted into kinetic energy $E_{k(\max)}$ for the electron. This relationship can be represented as follows:

$$hf = \phi + E_{k(\max)} \quad \text{or} \quad E_{k(\max)} = hf - \phi \quad (1)$$

$E_{k(\max)}$ can also be expressed as $1/2mv^2$, where m is the mass of the electron and v is velocity of the electron.

If the light photon has an amount of energy higher than the binding force of the electrons of that material, then the electrons can be released from its atomic orbit and start flowing as a current. For a given material, this binding force can vary only within a small range.

Assuming that the energy of the photon in incident light is sufficiently high that it causes the photoelectron to start flowing from a material, then a simple biased collector can be used to collect these electrons and convert the flow of current into a voltage signal. This voltage signal can be used to measure the changes in the chemical state of any surface.

Given the mathematical relationships expressed above for this principle, it is possible to measure the kinetic energy of the emitted electron. Since the amount of energy of the incident photon of light is known from the frequency of light applied, it is possible to determine the nature of the material. By subtracting the measured kinetic energy of the ejected photoelectrons from the known energy of the photon of light, it is possible to determine the work function of the material and thus determine the nature of the material and what may be deposited on the material.

The surface quality monitor (SQM) series instruments are based on the OSEE principle. When metals or certain other surfaces are illuminated with UV light with the proper wavelength (energy), electrons are emitted from the surface. The process by which UV photons interact with the surface to produce electrons is known as OSEE. The emitted and subsequently scattered electrons can be collected across an air gap by a biased collector (located at the end of the OSEE sensor, see [Fig. 3.1](#) for OSEE sensor illustration) and measured as a current which is then converted into a voltage and displayed on a digital panel meter (DPM) on the SQM series console. By maintaining the surface-to-collector distance relatively constant, changes in the measured photocurrent

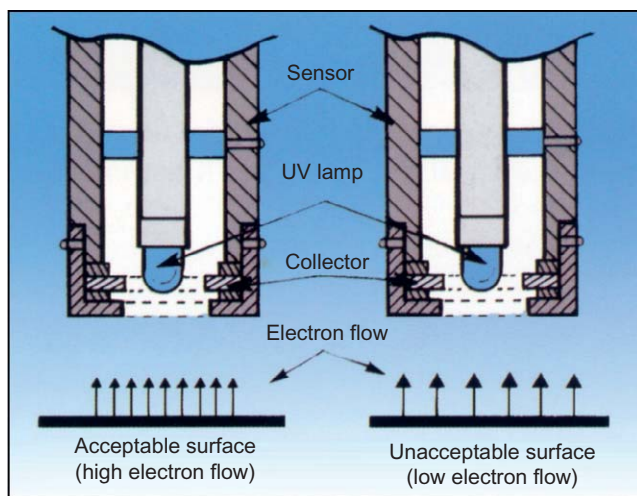


FIGURE 3.1 Cross section of an OSEE sensor.

(which is on the order of 10^{-10} to 10^{-12} A) can provide information about the surface, e.g., electronic structure and chemical composition. Any contaminant or thin film on the surface, depending on its own photoemission characteristics, can either enhance or attenuate the inherent emission from the clean surface. In simple electronic terms, the clean surface is a current generator and a non-photoemitting contaminant acts as a resistance because the current is attenuated by interactions between the electrons and the contaminants. The thicker the contaminant, the higher the resistance and, consequently, the greater the decrease in measured signal.

The OSEE sensor utilizes a low pressure mercury vapor lamp with two predominant peaks at 185 and 254 nm. These two wavelengths correspond to approximately 6.7 and 5.0 eV energy levels, respectively. The UV light flux provides the energy needed by the material so the electrons can emit (escape) from the surface.

The materials/surfaces that emit electrons when exposed to this UV light are called photoemitting materials and the other materials are called non-photoemitting. Both the substrate and the contaminant/coating can be emitting. A surface with no contaminant or coating on it usually gives a high signal. A contaminant or thin film, depending on its own emission characteristics, can either attenuate or enhance the signal from a clean substrate.

The presence of thin film, contaminant, or coating on the surface has two effects. First, it reduces the UV light reaching the substrate and second, it attenuates the flow of electrons away from the surface. Within a certain range, the signal attenuation is exponentially proportional to thickness of the thin film on the surface. Hence, the OSEE technique can also be used to measure the

thickness of thin films. The range of thickness measurement depends on the emission level of the substrate and the attenuation level of the contaminant/coating.

This mathematical relationship can be simplified and expressed as follows:

$$I_p = I_o \exp(-d/L) \quad (2)$$

where I_p is the OSEE signal with attenuating film of thickness d present on the surface. I_o , a constant, is the OSEE signal at $d=0$, that depends on the work function of the substrate. L , a constant, is the attenuation index representing the resistance to the electron/UV transport through the film. Equation (2) can be simplified to

$$d = \text{LN}(I_o)L - \text{LN}(I_p)L \quad (3)$$

LN is the natural logarithm.

2.1 Resolution

OSEE is extremely sensitive to very low level changes in surface chemistry. The technique is capable of detecting presence of a partial monolayer of thin film contamination.

Spatial resolution is approximately 10% of the area of the OSEE sensor opening. For example, if the OSEE sensor opening area is 6.5 cm^2 , the OSEE sensor should be able to detect 0.65 cm^2 of contaminated area within the 6.5 cm^2 area under the OSEE sensor opening. The actual spatial resolution will depend on the substrate, the OSEE sensor opening, and the thickness of contaminant on the surface.

OSEE is not capable of detecting or resolving particulate contamination, but is very sensitive to thin film molecular contamination.

2.2 Repeatability and Reproducibility

Figure 3.2 shows typical curves of OSEE versus time when a surface is exposed to UV light in air. The time and OSEE amplitude scales are usually different for different materials, but the shape of the curve is typical. The following discussion is therefore relevant to OSEE of any material.

When a surface is exposed to UV light, the OSEE signal increases very quickly to a peak signal, then the signal starts decreasing until it stabilizes. The solid line in Fig. 3.2 shows the signal when a surface is first exposed to UV light. The dotted line shows the signal when the same spot, on the same material, is exposed to UV light the second time. As can be seen, the second curve has a lower OSEE peak than the first curve. This phenomenon occurs due to a very complex process. An oversimplified explanation is that there is a charge that builds up on the surface. This charge will slowly dissipate after

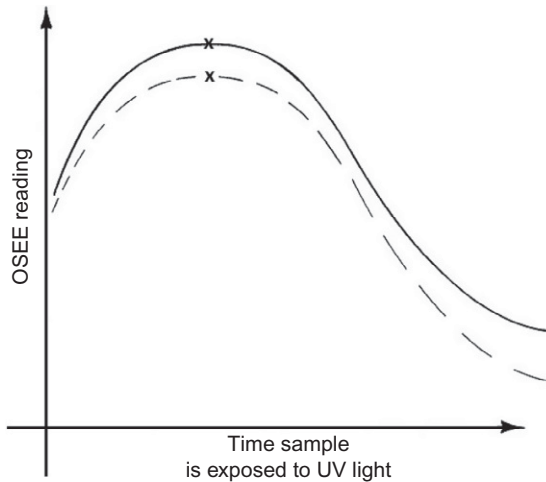


FIGURE 3.2 OSEE signal as a function of time of exposure to UV light.

the OSEE sensor is removed from the surface. This dissipation time varies with materials, ranging anywhere from a few minutes to many hours.

The conventional approach of repeated measurements on the same spot to show repeatability and reproducibility is not appropriate for OSEE. If measurements are made on grade 316 stainless steel, then the repeatability of measurement is much better than any other material. If a sample is scanned with consistent speed, then the OSEE repeatability and reproducibility can be demonstrated. For further information, see [Section 2.4.1.1](#).

2.3 Calibration

Until now, defining the level of cleanliness of any surface has been very subjective at best. OSEE provides the capability, for the first time, of being able to quantify surface cleanliness level. Since the idea of quantifying surface cleanliness is relatively new, there is no national standard on surface cleanliness. It is also not possible to have a calibrated reference for cleanliness measurement due to the difficulty in preserving the level of surface cleanliness of the calibration reference standard over a period of time.

Several direct calibration methods have been tried with limited success due to the difficulty of being able to maintain a clean surface. Calibration surfaces that have been tried are gold foil; chemically vapor-deposited gold on glass; chemically vapor-deposited chromium on glass, stainless steel, and chromium-plated steel. Of these, stainless steel was the most stable and reproducible; however, it was deemed to be not sufficiently stable to qualify as a calibration standard.

Hence, the system calibration is performed by using an indirect method. The indirect method involves two steps. The two factors that affect the calibration of the system over a period of time are: changes in the electronics, if any, and the natural aging of the UV lamp. The first step in indirect calibration is to monitor the intensity of UV light from the lamp and make adjustments (available only on models with optional high voltage, high frequency power supply) to keep the UV intensity constant over time. If no adjustment is available, then the readings should be adjusted to compensate for the percentage loss in UV lamp intensity. The intensity should be monitored using a research type radiometer.

The second step in indirect calibration involves checking the electronics using a calibration load (available from Photo Emission Tech.) and making necessary equipment adjustments. If no adjustments are available, then the readings should be adjusted to compensate for the percentage loss in the electronics calibration.

A calibration interval of 1 year (6 months for exacting applications) is recommended. Photo Emission Tech. also provides the calibration service.

2.4 Factors Affecting OSEE

There are a number of factors that, theoretically, can affect OSEE. The following discussion is intended to bring awareness of the effects of these factors. *It should be strongly emphasized that most of these factors have very little effect on the practical use of this technique for the intended applications including establishing a quantitative bench mark for surface cleanliness and using the technique for the ongoing monitoring of the cleaning process.*

These factors can be categorized into short-term and long-term factors. The short-term factors can be controlled relatively easily. The long-term factors can be monitored and, if necessary, can be corrected through calibration and/or replacement of component parts in the OSEE sensor. Each factor will be discussed along with its effect on OSEE reading and the type of action needed to control and/or mitigate the affect.

2.4.1 Short-Term Factors

2.4.1.1 Length of Time the Sample Surface is Exposed to UV Light

The rate of emission of electrons from most materials is much greater at the beginning of the exposure to the short wavelength UV light than after the surface has been exposed to the short wavelength UV light for some period of time. This effect is usually known as photoelectric fatigue [5]. The cause of this phenomenon has not yet been thoroughly determined, as it appears to depend on a variety of conditions, such as the nature of the substrate, the nature of the gas surrounding the substrate, the pressure of the gas, and even the quality of the UV light [5].

Figure 3.2 shows the effect on the OSEE signal of the length of exposure time to short wavelength UV light. The shape of the curve in Fig. 3.2 is a typical shape for all materials. The scales for both axes may vary with different materials. As noted earlier, the solid line represents the typical curve when the surface is exposed to UV light for the first time. The dotted line represents a typical curve when the same spot on the same surface is exposed to UV light for the second time. The points marked with an “X” are referred to as the peak readings. Each successive exposure of the surface to UV light during a short period of time (varies with material) will usually result in a lower peak reading.

All materials exposed to UV light eventually recover, i.e., the original reading can be repeated. The recovery time varies from material to material. It can be a few minutes (e.g., grade 316 stainless steel alloys) or a few hours (e.g., silicon wafers). No formal data are available on the recovery times for various materials. Even though all materials recover, it is seldom possible to repeat the original reading. This is due to the fact that, in most cases, in the time it takes for a material to recover, the surface of the material changes due to the effect of the environment, e.g., oxidation or contamination from the environment. In order to avoid changes over time due to environmental exposure, the sample should be stored and measured in an inert environment.

It has been shown that the use of argon gas, introduced through a small nozzle attached to the UV OSEE sensor and directed between the UV source/collector and a computer hard disk surface, controls the degree of decay [5]. The UVSEE signal can be made to remain constant on a stationary surface by adjusting the flow rate near 75 mL/min (2.65×10^{-3} scfm) and directing the flow relative to the OSEE sensor geometry. A beneficial byproduct of using argon is an increase of the detected electron surface signal (5-10%) without adding distortion to the same signal measured without argon.

When Winchester-type media (computer hard disk) with lubricant thicknesses of 1.4-3.6 nm were measured at Photo Acoustic Technologies on an OP1030 Disk Analyzer, an OSEE system particularly suited to measuring lubricant thickness on rigid disk media, the change in signal with time was absent due to the disk spinning at 500 rpm [6]. Since any spot on the disk was not under the OSEE sensor for an appreciable amount of time, the effect of decay in OSEE signal was absent and the resultant measurement was repeated over several scans of the disk. It was also shown that the lubricant thickness measurement using OSEE correlated extremely well with both electron spectroscopy for chemical analysis (ESCA) and Fourier transform infrared (FTIR) measurements on the same disks.

It should be noted that due to this unusual feature of the technique, conventional approaches to repeatability and reproducibility are not appropriate. See Section 2.2 for more information.

2.4.1.2 Distance of the OSEE Sensor from the Surface

Figure 3.3 shows the effect of the OSEE sensor distance from the surface on the OSEE readings. The closer the OSEE sensor is to the surface being inspected, the higher the OSEE signal. This change in OSEE signal is due to three factors. First, the emitted electrons have more distance to travel. Second, due to the collector being farther away from the surface, there is lower electric field. Third, the light flux decreases because of divergence of light.

Once a desirable OSEE sensor distance is established, as a rule, it should be maintained within ± 0.127 mm.

2.4.1.3 Line Voltage

Figure 3.4 shows the effect of variation in-line voltage, normalized for 110 V, on the OSEE readings. This curve is typical for all SQM series systems that have a high voltage transformer to power the UV lamp. In most applications, the line voltage is not a major problem. Line voltage may become a problem only in applications where very precise measurements are required, such as lubricant thickness within a few angstroms.

In those situations, where it becomes a problem, there are two possible solutions. One solution is to install a line voltage regulator. Photo Emission Tech., Inc.

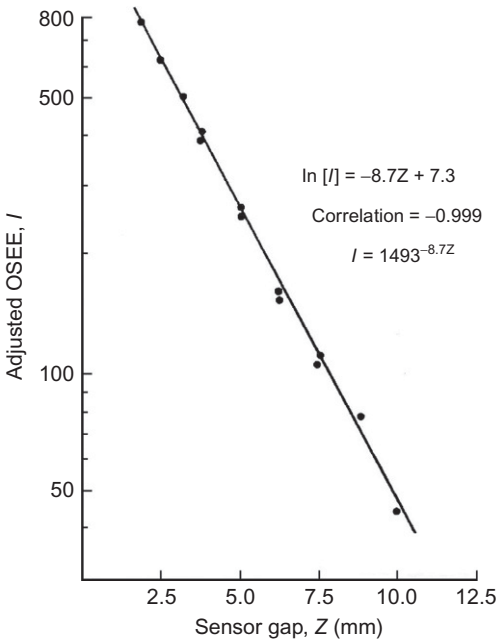


FIGURE 3.3 Adjusted OSEE signal versus distance of the sensor from the surface for a typical 25 mm sensor. The scale of “Adjusted OSEE, I” on the y-axis is in OSEE units or in centivolts. For example, a reading of 800 represents 8 V signal.

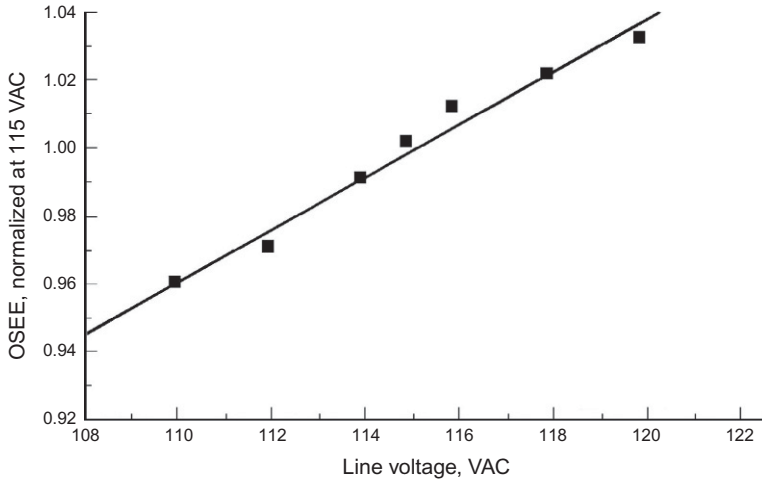


FIGURE 3.4 Normalized OSEE signal versus line voltage.

sells a regulator with $\pm 0.1\%$ regulation. The other solution is to buy SQM series equipment with a high frequency power supply. The high frequency power supply maintains a constant current to the lamp with a line voltage variation of $\pm 15\%$.

2.4.1.4 Ambient Temperature

In general, OSEE readings increase as the temperature increases. However, the percentage change in OSEE readings is much smaller than the percentage change in temperature. The relationship between OSEE and temperature is a very complex relationship and very hard to describe theoretically; however, experience shows that for most applications even a large change in temperature has very little effect on the OSEE.

2.4.1.5 Atmospheric Pressure

OSEE reading decreases as the pressure increases (see Eq. (4)). The exact relationship is not known. Some preliminary work has been done at NASA. The results indicate a very small change in OSEE for normal change in atmospheric pressure during a year.

$$\text{OSEE} = F\left(\frac{K}{P}\right) \quad (4)$$

Here, F represents a function, K is a constant, and P is absolute pressure = 0.101 MPa (760 torr). As P decreases, VD increases where VD is electron drift velocity. As VD increases, OSEE increases.

2.4.1.6 Sample Not Properly Grounded

It is important to always ground the samples that are to be tested, especially if the sample material is nonmetallic. The OSEE current is extremely low and can leak easily unless the sample is properly grounded. In general, it is recommended that all samples be grounded, although for most metallic materials testing without grounding the samples generally results in good measurements. The sample can be grounded by either attaching a conducting cable directly to the sample or to a metal (conducting) plate (then placing the sample on the grounded plate). The other end of this grounding cable should be attached to the OSEE system to complete the grounding of the samples.

In the case of some materials that are prone to static charge buildup, grounding may not be sufficient. It may be necessary to use some means of eliminating the static charge. For more information, see [Section 2.4.1.10](#).

2.4.1.7 Surface Finish

The OSEE reading changes as the surface finish changes. In general, the rougher the surface is, the lower the OSEE reading is for a given material. However, the change in OSEE readings, when compared to the change in signal due to contamination, is negligible for the range of surface finish/roughness normally encountered in the industry. Hence, for most applications, surface finish does not become an issue. [Table 3.1](#) lists data showing some finishes and corresponding OSEE readings.

2.4.1.8 Air Flow/Turbulence Past the OSEE Sensor

In general, the OSEE signal is higher and more stable when the air is still. The higher the flow of air is between the OSEE sensor and the sample, the lower are the OSEE readings. If the airflow is constant, i.e., not varying or turbulent, then the OSEE is constant assuming everything else is constant. If the equipment is used in an area where airflow is turbulent or varies, it will affect the OSEE readings. In situations where this may be a problem, it can be overcome by placing a shield around the OSEE sensor, or relocating the system to another area.

2.4.1.9 Humidity

The OSEE readings are affected by humidity. The readings increase as the humidity increases until the humidity becomes high enough that the water vapor condenses on the surface of the sample. In this case, the OSEE reading will be lower due to the water film on the surface. Data to determine how much effect the humidity has on OSEE needs to be obtained. It should be noted that the effect of humidity encountered in the industrial environment over a period of time has not been a problem in over 400 systems currently in use in the field.

TABLE 3.1 Effect of Surface Finish on OSEE [7]

Surface Roughness Height (in.) ^a	Relative OSEE Sensor Output
8 Ground with periphery of wheel	658
16 Ground with periphery of wheel	700
32 Ground with periphery of wheel	688
2 Lapped	870
4 Lapped	835
8 Lapped	750
63 Milled	800
125 Milled	824
250 Milled	640
63 Profiled	863
125 Profiled	799
250 Profiled	520

^aMeasured on a GAR 22 microfinish comparator (GAR Electroforming Division, Danbury, CT).

2.4.1.10 Static Charge

Static charge buildup on surfaces, particularly on nonmetallic, nonconducting surfaces, has an effect on the OSEE readings. The readings are erratic. These erratic readings can occur even though conventional static charge meters may not detect any significant charge.

A quick and convenient way to determine if static charge is a problem is to use the following procedure.

- (A) Set up the system to commence testing of samples in question.
- (B) Turn the peak detect on.
- (C) Place the OSEE sensor directly on the surface of the sample. *Note:* make sure that the sample is placed on a grounded surface.
- (D) Let the peak detect capture the peak reading on the DPM.
- (E) While observing the DPM, quickly remove the OSEE sensor away from the surface.
- (F) If the peak reading on the DPM increases, then static charge is a problem.
- (G) In the event static charge is a problem, we recommend the use of a static eliminating bar available through most dealers of static prevention equipment.

The materials known to pose a static charge problem are some plastics, such as Mylar and polyester, and glass.

2.4.2 Long-Term Factors

2.4.2.1 UV Light Intensity

The intensity of the UV lamp degrades with age and usage. The manufacturer specifies a half-life of 5000 h. This means that the intensity of the lamp will reduce to 50% of its initial intensity after 5000 h of use. As the lamp intensity degrades, the OSEE readings will drop, assuming all other factors remain constant.

The manufacturer also specifies a 4 h loss in life for each cold start of the lamp. Hence, if the lamp is going to be turned off for less than 4 h between uses, it is better to leave the lamp on. To assure consistent operation, the system should be calibrated periodically. See [Section 2.3](#) for additional information.

2.4.2.2 Collector Bias Voltage

Each OSEE sensor head has some batteries that are used to bias the collector. The voltage of the batteries is fixed. The batteries have very little drain on them; hence, the useful life of the batteries is the same as their shelf life. Typically, batteries life is 5-6 years. The batteries do not deteriorate; hence, their efficiency is not affected. Batteries keep functioning with the same efficiency throughout their life. They stop functioning suddenly at the end of their life. An easy way of testing the batteries is to touch the OSEE sensor collector with a finger. The reading on the DPM should go to approximately 1000. If there is no change in reading, then the batteries are depleted and need to be replaced.

3 PHOTOEMITTING MATERIALS

The materials that emit are those which have a work function of 6.7 eV or less. The lower the work function of a material, the more readily it will emit and the higher the emission level will be. Work function of a material is a measure of the amount of energy that is needed to release an electron from the surface. Both conducting and nonconducting materials can be photoemitting.

[Table 3.2](#) gives a partial listing of work functions of some materials. A more comprehensive list of work functions may be found in a handbook of physics.

Since both the substrates and the contaminants/coatings can be emitting or nonemitting, each case will be discussed below.

3.1 Substrate Emitting and Contaminant Nonemitting

[Figure 3.5](#) shows the theoretical relationship between the OSEE signal and contaminant/coating thickness for this case.

When the natural log of the OSEE signal is plotted against the thickness of contaminant/coating, the relationship is *inverse and linear* within a certain range of thickness. Within this range, OSEE is very useful for measuring the

TABLE 3.2 Electron Work Function (eV) of the Elements [7]

Element	Photoelectric Work Function (eV)
Ag	3.67
Al	4.08
As	5.11
Au	4.82
B	4.50
Ba	2.49
Be	3.92
Bi	4.25
C	4.34
Ca	2.76
Cb (or Nb)	3.96
Cd	4.07
Ce	2.84
Co	4.37
Cr	4.37
Cs	1.96
Cu	4.86
Fe	4.40
Ga	3.80
Ge	4.29
Hf	3.53
Hg	4.52
Ir	4.57
K	2.24
La	3.30
Li	2.28
Mg	3.68
Mn	3.76

Continued

TABLE 3.2 Electron Work Function (eV) of the Elements [7]—Cont'd	
Element	Photoelectric Work Function (eV)
Mo	4.34
Na	2.28
Nd	3.30
Ni	3.67
Os	4.55
Pb	3.97
Pd	4.97
Pr	2.70
Pt	4.09
Rb	2.09
Re	5.00
Rh	4.57
Ru	4.52
Sb	4.01
Si	4.2
Sm	3.20
Sn	3.62
Sr	2.74
Ta	4.13
Te	4.76
Th	3.57
Ti	4.17
Tl	3.68
U	3.63
V	3.77
W	4.35
Zn	3.89
Zr	3.73

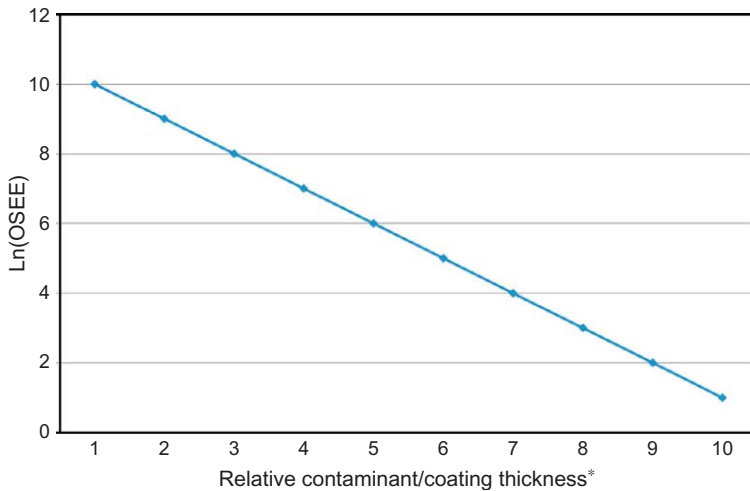


FIGURE 3.5 Theoretical relationship between the natural log of the OSEE signal and the contaminant/coating thickness. The substrate is emitting and contaminant/coating is nonemitting. (*The x -axis units of thickness are relative units. These units may be nm or mg/ft^2 or $\mu\text{g}/\text{cm}^2$ or any other thickness units.)

thickness of the coating/contamination. Beyond this thickness range, OSEE is very useful for detecting absence/presence of contamination/coatings.

3.2 Substrate Nonemitting and Contaminant Emitting

Figure 3.6 shows the theoretical relationship between the OSEE signal and contaminant/coating thickness for this case.

When the natural log of the OSEE signal is plotted against the thickness of contaminant/coating, the relationship is *direct and linear* within a certain range of thickness. Beyond this thickness range, OSEE has maximum sensitivity to detect absence/presence of contamination/coatings.

3.3 Both Substrate and Contaminant Emitting

Figure 3.7 shows the theoretical relationship between the OSEE signal and contaminant/coating thickness for this case.

If the substrate is much more emitting than the contaminant/coating, then the relationship between the natural log of the OSEE signal and the natural log of thickness is *inverse and linear*, as shown by the red curve labeled “substrate emitting.”

When substrate is much lower emitting than the contaminant/coating, then the relationship between the natural log of the OSEE signal and the natural log of thickness is *direct and linear*, as shown by the blue curve labeled “contaminant/coating emitting.”

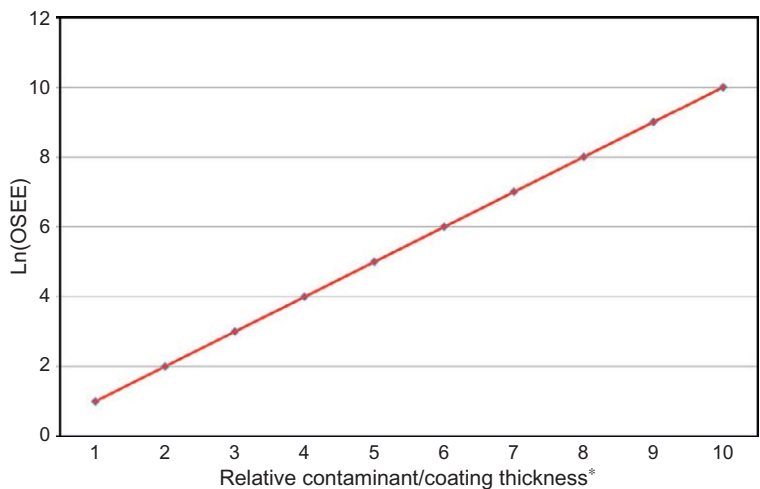


FIGURE 3.6 Theoretical relationship between the natural log of the OSEE signal and the contaminant/coating thickness. The substrate is nonemitting and the contaminant/coating is emitting. (*The x -axis units of thickness are relative units. These units may be nm or mg/ft^2 or $\mu\text{g}/\text{cm}^2$ or any other thickness units.)

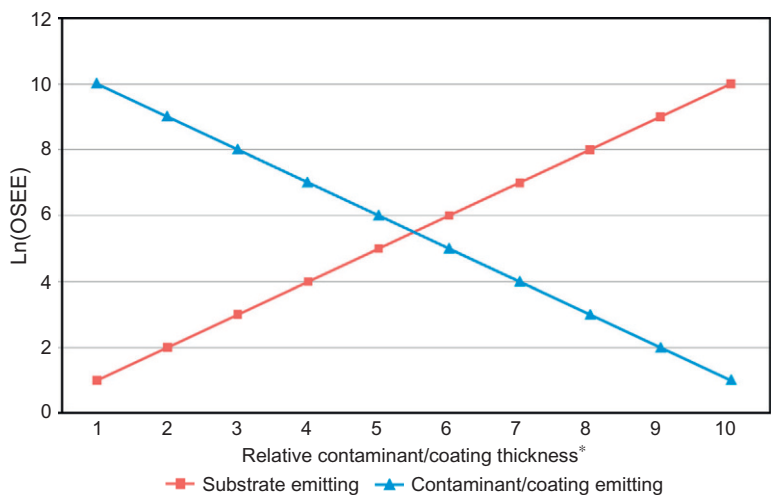


FIGURE 3.7 Theoretical relationship between the natural log of the OSEE signal and the contaminant/coating thickness. The substrate and contaminant/coating are both emitting. (*The x -axis units of thickness are relative units. These units may be nm or mg/ft^2 or $\mu\text{g}/\text{cm}^2$ or any other thickness units.)

3.4 Both Substrate and Contaminant Nonemitting

In this case, the OSEE cannot be used.

4 APPLICATIONS OF OSEE

OSEE is extremely sensitive to the changes in the chemical/metallurgical state of a surface. In general, therefore, OSEE can be a very useful technique when changes in the chemical/metallurgical state of a surface need to be detected and monitored.

The following are some specific areas of application for OSEE. Each area of application is discussed in detail with specific examples.

- (1) Surface cleanliness monitoring.
- (2) Detecting absence/presence of films/coatings.
- (3) Measuring thickness of thin films/coatings.

4.1 Surface Cleanliness Monitoring

A quality surface finish requires a surface that is “properly” prepared. The preparation may involve absence of thin film contamination and/or presence of certain type and thickness of a coating (pretreatment).

Contamination is defined as any unwanted or foreign material on surface. Defining and maintaining the surface preparation at “proper” levels is, at best, subjective. Often the failure of the surface preparation processes is not discovered until poor quality surface finish occurs, such as poor adhesion of paint/coating.

Considerable empirical work has been done to study and establish suitable surface preparation processes. These processes are controlled by monitoring various characteristics of the process, such as temperature and chemical concentration levels. Despite stringent controls, poor quality product is generated for various reasons. The reasons include, but are not limited to, the number of parts processed between cleaning chemical replenishment and the variation in the amount of contamination on each part.

Unlike some other techniques, OSEE is a direct technique, i.e., the system can be used to make measurements directly on the surface of interest. In addition, OSEE is a quick technique, thus it is possible to use this technique on-line for real-time measurements. It is a nondestructive, noncontact method to help establish a quantitative measure of surface cleanliness/preparation. It can be extremely helpful in optimizing surface preparation processes by providing quantitative information about the effect on surface cleanliness/preparation for a given change in the process. The technique can be used very effectively to assure that the process is under control.

4.1.1 *Establishing Surface Cleanliness Level*

4.1.1.1 **Selecting an Appropriate Cleanliness Monitoring Technique**

A key to surface finishing is a “properly” prepared surface. Defining and maintaining the surface preparation at “proper” levels is at best subjective. To assure consistent level of surface cleanliness, it is important not only to specify a desirable level of surface cleanliness but also to specify a method of cleanliness verification for that level to have meaning.

This generally leads to the question, what do we mean by “clean”? How clean is “clean”? Even so-called clean parts have a certain amount of contamination, even if it is at a microscopic level. With advances in technology, more and more applications are moving toward the need for higher levels of cleanliness. Hence it is important to select an appropriate surface cleanliness verification technique [8].

As noted in Ref. [8], “In order to define an acceptable level of cleanliness, it is important to minimize the total cost of cleaning. The total cost of cleaning is the sum of the cost of achieving a certain level of surface cleanliness and the cost of failure/nonconformance associated with that level of surface cleanliness. Selecting a method for monitoring cleanliness is the first step in establishing an acceptable level of cleanliness or defining ‘how clean is clean.’ Several factors need to be considered in selecting an appropriate surface cleanliness method, which include, but are not limited to, type of contaminant to be detected, level of cleanliness to be monitored, acquisition and operating cost of the monitoring method, and the skill level required to operate the system. Surface cleanliness monitoring method may be direct, indirect, or analytical. Monitoring method can be used to optimize the cleaning process by varying different parameters of the cleaning process while monitoring surface cleanliness to see how it is affected by the change. It can also help in ongoing monitoring of the cleaning process to assure that the desirable level of cleanliness is being achieved” [8].

The first criterion that needs to be considered is how large an area should be monitored on a sample. The factors effecting this decision include the spatial resolution required, knowledge of the process, and throughput required. In general, the more the spatial resolution is required, the smaller the area of inspection has to be. Typically, for OSEE, the spatial resolution is 10% of the area inspected. If the OSEE sensor opening area is 6.5 cm^2 , then a contaminated area of 0.65 cm^2 , within the 6.5 cm^2 area being inspected, can be detected.

The second criterion is to determine whether the test is to be static or dynamic. A static test is where the part is stationary when the test is done. A dynamic test is where the part is moving past the OSEE sensor in some material flow process. In a static case, it is recommended that the OSEE sensor be placed on the sample surface and “peak detect” mode on the instrument be used to capture and record the highest reading from the surface. If a noncontact test is desired, then the OSEE sensor must be mounted in a stand and the part placed under the OSEE sensor. In a dynamic test, the OSEE sensor is mounted such that

the OSEE sensor distance from the sample surface is as small as possible (less than 6.35 mm, typically 3.18 mm) and this distance is maintained.

The third criterion is to determine if data recording/archiving capability is needed. This feature is extremely useful if an independent process validation is a requirement for the application on hand. It is also useful for historical purposes to be able to review the cleanliness data with a view to improving the process and reducing nonconformance due to poor surface cleanliness. Model SQM300 automatically records all data for future review.

4.1.1.2 Establishing Acceptable Level of Cleanliness

There are two approaches to establishing an acceptable level of cleanliness: empirical approach or quantitative approach. The empirical approach involves ascertaining the level of cleanliness achieved by the current cleaning process. The quantitative approach can be used in an application, where a subsequent operation is required that is dependent on surface cleanliness and whose success can be quantified and measured, such as adhesion of a coating or bonding of parts.

4.1.1.2.1 Empirical Approach In order to use the empirical approach, a statistically significant sample (minimum of five samples) of clean parts should be taken. Each part should be tested and the peak OSEE reading recorded. Once the readings from all the samples have been recorded, the mean and the standard deviation of these readings should be calculated.

It is recommended that ± 3 standard deviation limits be established as the upper and lower acceptance level for the surface cleanliness. These limits can be set on the OSEE instrument as upper and lower gate settings. Once these settings are established, production parts can be tested. If the cleanliness of the parts is outside the acceptance range, the red light on the OSEE sensor and on the front panel of the control unit will come on indicating rejection; otherwise, the green light will serve to indicate the part is clean within the specification.

Once a quantitative acceptance criterion has been established, the cleanliness of the part can be monitored on a sampling basis. The parts can be followed downstream to make sure that the nonconformance due to surface cleanliness is acceptable. If the nonconformance due to surface cleanliness is not acceptable, then the level of surface cleanliness should be increased until the downstream nonconformance level is acceptable.

4.1.1.2.2 Quantitative Approach In order to use the quantitative approach for establishing an acceptable level of cleanliness, the success of the downstream operation that relies on good surface cleanliness should be defined in quantitative terms. For example, if parts are to be bonded together, then an acceptable level of bond strength needs to be defined.

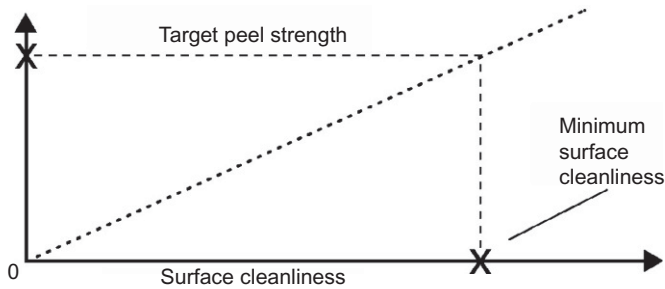


FIGURE 3.8 Peel (bond) strength versus level of cleanliness.

A statistically significant set of samples (minimum of five samples for each cleanliness level) should then be cleaned. The samples should be contaminated with a known quantity of the most common contaminant encountered in the normal process. The range of contamination levels should be broad enough to cover the expected level of surface cleanliness needed to achieve an acceptable level of bond strength. Several OSEE peak readings from each part should be taken and recorded. Once the readings from all the samples have been recorded, the mean and the standard deviation of these readings should be calculated.

Next, the samples should be processed in the same manner as the normal production parts for bonding. After the bonding is done, then all the samples should be tested for bond strength. The bond strength measurements should be recorded and correlated with the mean OSEE readings for each set of samples. If the results of bond strength are plotted against the level of cleanliness (mean OSEE readings) for the group of samples, the result would be linear correlation between the bond strength and surface cleanliness as shown in Fig. 3.8.

The minimum acceptable level of cleanliness would be the one that correlates to the target peel (bond) strength. Once a quantitative acceptance criterion has been established, then the part cleanliness can be monitored on a sampling basis. The parts can then be followed downstream to make sure that the target peel (bond) strength is achieved. If the target peel (bond) strength is not achieved, then the level of surface cleanliness should be increased until the target peel (bond) strength is not achieved. For further information, see Application 1 in the following Section 4.1.2.1.

4.1.2 Example Applications of Surface Cleanliness Monitoring

4.1.2.1 Prebond Surface Quality

4.1.2.1.1 Application 1 As noted in Ref. [7], “The performance of many critical components for the Space Shuttle and other flight hardware depends on the quality of bonding achieved during manufacturing. An example of a major shuttle element, where quality bonding is crucial to performance reliability and safety, is the Solid Rocket Motor (SRM). Inadequate bonding of the

rubber insulation to the case could result in exposure of the D6AC steel case to the hot gases from the burning propellant and result in a burn-thru which could be disastrous. Thin film surface contamination is probably the most insidious and least understood parameter that affects bonding” [7].

Studies were performed at NASA Marshall Space Center in Huntsville, AL to correlate the relationship between the levels of contamination versus the peel strength of the bond. The relationship between the levels of contamination versus the OSEE signal was also established.

Prior to the use of OSEE, there was no method to detect contamination levels less than 100 mg/0.1 m², which resulted in average bond strength of 1054 kg/m {59 pounds per linear inch (PLI)}. The use of OSEE gave NASA the capability of monitoring surface contamination down to 0.25 mg/0.1 m². Better cleaning methods were developed by using OSEE to optimize the cleaning process. As a result, an average bond strength of 3500 kg/m (196 PLI) was achieved, a three-fold increase in bond strength.

4.1.2.1.2 Application 2 The reliability of the carbon fiber-reinforced plastics (CFRPs) is governed by assuring the surface quality of CFRP material. Roughness of ready-to-bond CFRP surfaces affects, in a complex way, their wetting behavior as determined by water break test [9].

OSEE was investigated as an in-line technique to assess the surface quality prior to bonding. Sensitivity and accuracy of OSEE measurements were successful in distinguishing favorable surface states of CFRP adherents from surface states which were unfavorable for adhesive bonding. It has been shown that OSEE can be applied in the field and without electrically contacting the CFRP surface for sensing moisture, thermal degradation, or thin layers of contaminants such as release agent and hydraulic oil for aeronautical use.

4.1.2.1.3 Application 3 As stated in Ref. [9], “If a fiber-reinforced composite structure is to be assembled by adhesive bonding, it is necessary to have some assurance that a good bond will be formed. When bonding metals (particularly aluminum) it is relatively easy to obtain good initial bond strength but less easy to ensure durability in hot and humid conditions. With epoxy matrix carbon fiber composite, once a good initial bond to an epoxy adhesive has been formed there is no subsequent loss of strength due to degradation of the adhesive-composite bond. The problem lies in ensuring a ‘clean’ surface initially.

“Surfaces were examined using X-ray photoelectron spectroscopy (XPS) analysis and OSEE. A reasonable correlation was found between the OSEE signal and the XPS analysis when the latter was taken in the form of estimated contaminant coverage or as the nitrogen content on the surface.

“There was a strong relationship between OSEE signals and adhesively bonded joint strengths, both lap joint and peel, and with the treatment of the composite surface before bonding. In order to ensure high joint strengths and low proportions of failure at the adhesive/composite interface, an OSEE signal

of ~60% of the maximum obtainable from the cleanest possible composite surface is required, a value that agrees with that obtained from the comparison of XPS and OSEE data. While further work is required to characterize it more fully, UV photoelectron emission does appear to be a possible method for the nondestructive quality control of surfaces of epoxy resin matrix composites before adhesive bonding" [9].

"The implementation of CFRP as a structural material in the aircraft industry requires a reliable quality assurance of surface states for adhesive bonding techniques. Improper grinding during mechanical surface pretreatment, contaminants on adherents, moisture uptake in the CFRP bulk, or thermal impact can have a large negative influence on the mechanical properties of adhesive joints prepared in manufacturing or repair processes. Therefore, assurance of surface quality needs to be implemented in order to obtain the best conditions for an adhesive joint.

"The optimum CFRP surface state before adhesive bonding denoted as UT had been obtained after grinding until exposure of a carbon fiber layer. It was characterized with respect to its structure and composition using vacuum-bound analyses. Applying OSEE as a surface-sensitive and in-line-capable inspection tool operated in ambient atmosphere, these CFRP surfaces showed prominent OSEE signal intensities which were obtained without electrically contacting the CFRP material.

"With OSEE signal intensities being sensitive to the presence of adsorbents on top of such photo-stimulated CFRP surfaces and possibly also to the electric properties of fiber/matrix interphases, monitoring by OSEE showed its capability to distinguish all the investigated surface states known to be unfavorable for adhesive bonding from the CFRP surface state UT. In detail, four types of unfavorable conditions of CFRP panel surfaces were considered. Thermally impacted adherents could be distinguished as well as moist adherents and adherents covered with release agent or hydraulic oil for aeronautical use.

"In conclusion, the OSEE technique demonstrated its potential for surface state quality assurance of CFRP materials prior to adhesive bonding" [9].

4.1.2.2 Surface Finish

In one application, aluminum panels were anodized, silk-screened, sheared, and shape-formed for control panels and decals for household appliances and automotive trim. The OSEE technique was used to establish the acceptable level of cleanliness to improve the adhesion of anodizing and silk-screening to the aluminum panel. The effectiveness of the cleaning process was also monitored in-line to assure that the surface cleanliness met the established standard.

In another application, steel sheet was hot dip galvanized. There was a problem of "bare spots," areas on the steel sheet which had no adhesion of the galvanized coating. This problem was minimized by using OSEE to establish

and monitor, in-line, the effectiveness of the cleaning process prior to coating. An additional benefit of the use of this technique was to reduce the chemical concentration of the cleaning bath by 40% and still achieve an acceptable level of cleanliness and improve adhesion of the coating.

4.1.2.3 Selecting the “Right” Cleaning Process

From Ref. [10], “Industry faces a dilemma in trying to select the ‘right’ cleaning process. Evaluating alternative cleaning processes requires the answer to two questions: (1) Is the new cleaning process as effective as the one being replaced? and (2) Does the new cleaning process leave behind any residue that could be detrimental to subsequent processing or performance?” [10].

OSEE is a simple-to-use technique for quantifying the level of surface cleanliness to help answer both of these questions. The technique can be used to: establish, quantitatively, the cleanliness level achieved by an existing cleaning process; measure the cleanliness level achieved by the alternative cleaning processes to help select the most effective and cost efficient alternative; and monitor the effectiveness of the cleaning process on an ongoing basis.

The OSEE technique can help answer such questions as: (1) effect of time elapsed after cleaning on the level of surface cleanliness, (2) evaluation of different cleaning processes for the best level of cleanliness, (3) evaluation of cleanliness levels for different levels of concentration of the cleaning agent and different processing times, and (4) effect of cleaning and cleaning time of different contaminants on surface cleanliness.

4.1.2.4 Optimizing a Cleaning Process

Generally, control of a cleaning process involves: (1) concentration level of the surfactant or cleaning agent, (2) temperature of the surfactant and cleaning agent, (3) the rate of flow of the surfactant or cleaning agent, (4) amount of time the part is cleaned, and (5) the time interval between replenishment or replacement of the surfactant and the cleaning agent. Most of these factors are generally established by the company supplying the cleaning process equipment and cleaning agents. OSEE can be used to establish a relationship between each of these factors and its effect on surface cleanliness. Studies can be performed by varying each of the first four factors and measuring their effects on surface cleanliness. Thus, the study of these factors can help “optimize” any cleaning process.

The time interval between replenishment or replacement of the surfactant and cleaning agent should not be fixed, since it can depend on two main factors. First, the amount of contamination can vary considerably from part to part or parts received from different vendors. Second, the number of parts being cleaned during the fixed replenishment/replacement time interval can also vary considerably due to production flows and ebbs. Thus, a fixed time-dependent replenishment schedule is not the best way of controlling the cleaning process.

Using OSEE for in-process monitoring of surface cleanliness to a preestablished “acceptable level” will help determine when the replenishment of chemicals or cleaning agents is necessary, and not done according to a predetermined, somewhat arbitrary schedule. The time interval between replenishment of chemicals would be dependent on the effectiveness of cleaning process and not on a fixed time interval.

4.1.2.5 Contamination Detection on Weld Surfaces

From Ref. [11], “Weld quality is largely dependent on the cleaning method. The cleanliness of some sensitive materials was not being verified. It was also difficult to determine the optimum cleaning process for materials prior to welding. In addition, a relative cleaning comparison is necessary to determine the effectiveness of different solvents/detergents. All this was needed to improve the cleaning processes used in production. OSEE technique was evaluated for its effectiveness to determine cleanliness of materials prior to welding, determine the relative cleaning capabilities of various processes, techniques, and cleaning media on different materials.”

“Contamination on metal surfaces to be welded is another area of considerable concern. Contamination in the weld bead increases porosity of the weld, with a reduction in weld strength. As a result of this concern, OSEE has also been applied in the inspection of surfaces prior to welding. In addition to inspection of Weldlite and other aluminum materials, various combinations of cleaning processes (solvents/detergents/methods) were evaluated by Martin Marietta Aerospace. A Taguchi experiment was utilized to establish the optimum cleaning method for contaminants common to the manufacturing process, e.g., marking pencil and surface oxidation. The result has been improved preweld surface preparation and the ability to implement a quality control procedure prior to welding” [11].

4.1.2.6 Copper Foil Characterization and Cleanliness Testing

From Ref. [12], “The ability to process inner layers with fine-circuit features can be influenced by characteristics of the copper foil used to start the process. The characteristic that has the most effect on adhesion is cleanability, which can be measured by a relatively simple instrumented procedure. A procedure based on OSEE was found to be efficient and reliable and should be considered as a tool for statistical process control of incoming material” [12].

During implementation of statistical process control (SPC) concepts in circuit inner layer manufacturing, it was noted that variations in copper foil characteristics could contribute to variations in process control and quality. Two critical characteristics were identified: (1) cleanability of the vendor-applied chromate passivation treatment, and (2) copper crystal orientation, which affects the sidewall geometry of circuit paths. Among screening methods evaluated for SPC of cleanability, a surface monitor based on the photoelectric effect was

found to be the most efficient and reliable. The effect of crystal orientation on etch performance affected only line circuits with a width of 76 μm or less, and only when the circuit features were generated under controlled process conditions.

The OSEE technique was found to be efficient and reliable for checking the cleanliness level of the copper foil on printed circuit boards.

4.1.2.7 Inspection for Flux Residue on Electronics Assemblies

With a new generation of fluxing, soldering, and cleaning technologies in the electronics industry, the entire assembly process needs to be reevaluated from a yield, quality, and reliability perspective. From the standpoint of long-term reliability, it is important to distinguish between innocent chemical residues and “dangerous” ones. Cleanliness detection methods, of course, change with the use of new solder fluxes and pastes. Several on-line detection methods popular in the rosin flux/CPC (Division of Premier Farnell Ltd., Leeds, UK) cleaning era have become questionable in the emerging technologies using “no-clean” or low solids flux soldering [13].

The OSEE technique was extended for inspection of insulating substrates used in printed wiring board construction by altering the measurement procedure to include charge replacement, thereby attaining measurement reproducibility. The results indicate that OSEE inspection of electronic assemblies for flux residues is feasible. An inspection based on this technology subjects the inspected object only to photons of UV light and immersion in an inert gas, such as argon. It is potentially rapid enough to provide 100% inspection of boards processed on a production line, and it has potential spatial resolution of less than 1 μm [13].

As stated in Ref. [13], “The feasibility of OSEE for inspection of electronic assemblies has been demonstrated. The detection limits associated with each method were established. In addition, the pros and cons of these methods as routine quality control inspection tools were discussed. OSEE was demonstrated to be a sensitive technique for detection of low levels of flux circuit residues on insulating substances. However, future work including development of rugged OSEE instrumentation will determine whether the electronics assembly industry can accept this technique in a production environment” [13].

4.1.2.8 Detecting of Fingerprints on Beryllium Mirrors

The utility of OSEE for inspection of fingerprints on polished beryllium surfaces was evaluated for an optical mirror application. In the manufacturing environment, oils/salts associated with fingerprints generally go unnoticed until they have etched or corroded the surface. OSEE easily detected the fingerprints on beryllium, whereas visual detection was not effective [14].

4.1.2.9 Graphite/Epoxy, Composite Skins on Honeycomb Panels

The potential use of OSEE was briefly assessed for graphite/epoxy skins on aluminum honeycomb-core sandwich panels. The exterior cleanliness of the skins was evaluated. Both sides of an aluminum honeycomb-core sandwich were inspected, then twice solvent (Freon) wiped and inspected. The results showed that the electron emission characteristics of the smoother surface (i.e., the side in contact with the mold) was increased by the solvent wipe, while there was no apparent change on the rough side (in contact with the vacuum bag). Presumably, some mold release had transferred to the surface and was removed by the solvent wipe. It was also noted that charge buildup occurs when using too slow a scan rate and is manifested by the display of erratic signals. This brief experiment suggests that some graphite/epoxy composite structures may be inspectable for contamination using OSEE [14].

4.1.2.10 Battery Vent Discs

From Ref. [14], “A test program was performed to evaluate the potential of OSEE to detect thionyl chloride vapor on 316 grade stainless steel, assess the utility of a sodium bicarbonate cleaning procedure, and evaluate the effectiveness of a nonmetallic coating (Heresite P403, Heresite Protective Coatings, Manitowoc, WI) for corrosion protection. The 316 grade stainless steel is utilized in lithium battery vent discs with a critical thickness of 12.7 μm . The lithium batteries are filled with the thionyl chloride electrolyte solution under very dry conditions because of the hygroscopic/corrosive nature of the thionyl chloride. If the electrolyte (liquid or vapor) is deposited on the surface (e.g., during filling) and is not removed, battery surface (including the vent disc) will rapidly corrode when moved from the dry environment to ambient humidity conditions. OSEE readily detected the presence of the electrolyte on the steel surface at levels that were not visible to the eye. In addition, the technique was extremely useful in establishing the suitability of the proposed surface cleaning process. On the other hand (not surprisingly), the OSEE was not effective in inspection of the Heresite P403 corrosion-protection coating. This is likely due to the nonmetallic nature of the Heresite coating. Unlike metals, electrons in non-metals are tightly bound, thereby reducing the concentration of electrons available” [14].

4.1.2.11 Solvent Rinse Cleanliness Verification

From Ref. [15], “The NASA White Sands Test Facility (WSTF) has an ongoing effort to reduce or eliminate usage of cleaning solvents such as CFC-113 and its replacements. These solvents are used in the final cleaning and cleanliness verification processes for flight and ground support hardware, especially for oxygen systems where organic contaminants can pose an ignition hazard. For the final cleanliness verification in the standard process, the equivalent of 0.1 m^2 of surface area of parts is rinsed with the solvent, and the final 100 mL of the rinse

is captured. The amount of nonvolatile residue (NVR) in the solvent is determined by weight after evaporation of the solvent. An improved process of sampling this rinse, developed at WSTF, requires evaporation of less than 2 mL of the solvent to make the cleanliness verification. Small amounts of the solvent are evaporated in a clean stainless steel cup, and the cleanliness of the stainless steel cup is measured by using a commercially available SQM. The effectiveness of this new cleanliness verification technique was compared to the accepted NVR sampling procedures. Testing with known contaminants in solution, such as hydraulic fluid, fluorinated lubricants, and cutting and lubricating oils, was performed to establish a correlation between amount in solution and the process response.

“Cleaning operations at WSTF encompass support for materials, components, propulsion test systems, and a sizeable shuttle depot activity. As part of these operations, components and piece parts are cleaned for a variety of service in gaseous and liquid environments including oxygen, hydrogen, hydrazines, and nitrogen tetroxide. Most of these parts are constructed of stainless steel and are resistant to corrosion in aqueous media. For precision cleaned parts, the standard cleaning process consists of two major elements: an aqueous precleaning process and a solvent-based final cleaning and cleanliness verification. Cleanliness levels are specified by a number that has an associated number of particles at this maximum size in micrometers followed by a letter to specify the maximum residue allowed on 0.1 m² of surface area. The letters A, B, C, and D refer to 1, 2, 3, and 4 mg of residue, respectively. Cleanliness levels to 50 A are required for high-pressure oxygen service. Following the aqueous precleaning, the parts are dried, visually inspected, and if passed, are transferred to the Class 100 clean room. The accepted process for solvent-based final cleaning is to rinse with HFE 7100 in the vapor degreaser followed immediately by a sampling rinse of approximately 100 mL/0.1 m² of component surface area. This rinse is then filtered for a particulate count and then submitted for NVR determination. The components are then blown dry with nitrogen, packaged, and labeled for use.

“First the solvent is transferred to a boiling flask, and the volume of the solvent is reduced to 10-15 mL by distillation. The solvent trapped in the distillation by a condenser is recycled the remaining 10-15 mL is transferred to a tared weighing pan and placed in a 100 °C oven for 30 min to complete the drying. The weighing pan is allowed to cool in a desiccator before weighing again to determine the NVR. For HFE 7100, the process takes approximately 40 min. It should also be noted that some of the more volatile contaminants could be volatilized, especially during the 30 min drying in the oven. Light hydrocarbons can show 40% per NVR recovery or less from spiked solvent samples.

“A method for cleanliness verification has been developed using the OSEE system (Model # SQM200). A number of contaminant responses were measured and the responses were very nearly equal. By sampling the rinse solvent two or more times, the confidence in the measurements is increased. Because of the

small amount of solvent used, retests are possible. The conclusion is that the SQM200 and NVR methods are equivalent in measuring surface cleanliness. The SQM offers significant savings in the time required to process the NVR. In handling and evaporating less solvent, significant solvent savings can be realized" [15].

4.1.2.12 Steel Surface Oxidation After Pickling

As noted in Ref. [16], "Oxide film, which is present at normal conditions on the steel surface, prevents the achievement of good adhesion between the substrate and protective metal coatings (zinc, nickel, copper, and so on), for example, electro- or electroless plating and hot dip coating of steel surfaces. Usually, oxides are removed by pickling, fluxing, or annealing in furnace in a vacuum. After pickling, steel surface is rinsed by water sprays or immersed in a tank with flowing water prior to transfer to the coating stage—plating or fluxing before hot dip coating. On high speed continuous strip, tubing, or wire lines, the rinse time does not exceed several seconds. If the line speed is not high enough, growth of oxide film can occur prior to coating. This is also true for batch processes if the time interval between exit from the pickling tank and immersing in a coating (fluxing) solution is too long. The most favorable conditions for rapid oxide film growth exist when the steel surface is spray rinsed. The growth of oxide film thickness can reach a substantial value in 10-15 s, in some cases even red rust can be seen on the steel surface" [16].

Oxidation of cold rolled steel panels after pickling in different acids, such as 30 s in 10% HCl, 10% solution of sulfuric acid and phosphoric acids, 2% HNO₃, and 10% HCl with 2 g/L of organic inhibitor and rinsing with tap water and in a beaker, was investigated by the OSEE technique. The steel surface, after pickling, rinsing, and blotting and then exposure to ambient air, forms an oxide film very quickly. OSEE was able to determine the level of cleanliness and differentiate the ability of various techniques to protect steel surface from oxide growth.

4.1.2.13 Performance Monitoring of an Aqueous Cleaner for Metal Surface Finishing Processes

Due to environmental regulations, the use of ozone depleting chemicals (ODCs) was being phased out. To meet this new regulation, Lockheed Missiles & Space Company started using an aqueous cleaner to replace 1,1,1 trichloroethane, an ODC, to prepare aluminum surface prior to adhesive bonding, conversion coating, or anodizing.

A vendor-recommended titration method was being used to monitor and control the aqueous cleaner. This titration method only measured the alkalinity of the bath. To better control the process, a method of measuring the performance of the bath was desired.

A two-part study evaluated three different techniques—weight gain, NVR, and OSEE—for measuring contamination and using these techniques to control the aqueous cleaning process [15].

The test results showed the OSEE technique to be the most sensitive at the lower levels of contamination. The level of contamination detected by OSEE could not be detected by either the weight gain method or the NVR method.

During the study of these techniques for controlling the aqueous cleaning process, it was found that the agitation of process solution did not enhance surface cleanliness, whereas the agitation of the rinse solution is critical to surface cleanliness. It was also concluded that the weight gain method was suitable for detecting contamination greater than $2 \text{ mg}/0.1 \text{ m}^2$, the OSEE method was most sensitive and could detect $1\text{--}10 \text{ mg}/0.1 \text{ m}^2$. The NVR method was the least desirable for controlling the aqueous cleaning process.

4.1.2.14 Other Applications

From Ref. [14], “While the OSEE technique is expected to be less effective for inspecting nonconductive, nonmetallic surfaces, there has been interest in evaluating its utility for certain nonmetallic materials used in space systems.

“An elastomeric material was inspected using OSEE to determine its use for verification of plasma-etch treatment prior to bonding. A comparison of OSEE readings for pre- and post-etch materials was performed. The OSEE readings for the post-etch sample were notably higher than those of the pre-etch material. This indicates that some elastomers (likely depending on fillers) are OSEE-inspectable. However, it should be noted that charge buildup occurred when inspecting at too slow a scan rate, resulting in erratic readings.

“Zinc orthotitanate (ZOT) is a silicate thermal-control coating, often applied to vapor-deposited aluminum on graphite/epoxy structures. ZOT is also utilized as a thermal-control coating on some space system radiator panels to provide low solar absorptance-to-emittance ratio (α_s/ϵ). Contaminants will cause the solar absorptivity to increase when exposed to space radiation, thereby degrading thermal-control performance. Common contaminants not visually detectable, such as pink pearl erasure and fingerprints, were detected by OSEE, albeit marginally.

“In addition, nonconductors such as second surface mirrors and fused silica (with a dielectric coating) were evaluated. Contaminants for this assessment were those common to personnel such as fingerprints and spittle; however, no significant response to OSEE inspection was detected.

“OSEE has demonstrated great utility as a noninvasive inspection technique for the detection of molecular contamination and is a considerable improvement over ‘black light’ inspection. The equipment can be modified and expanded to incorporate automation (especially useful for large surfaces). Preliminary efforts to obtain OSEE readings are required to characterize a new substrate and/or contaminant. OSEE can provide a quantitative measure of

contamination present, providing the contaminant is known. For unknown contaminants, the information would be represented as a deviation from the reading obtained for a clean bare substrate. OSEE cannot identify the nature of a contaminant.

“The OSEE technique is regarded as a useful tool for rapid, noninvasive inspection to determine deviations from a standard clean surface. It is especially applicable to manufacturing process controls for which surface conditions are critical. Examples include bonding, coating, and welding. It may also be applicable to optical and thermal-control surfaces wherein presence of contaminants would cause the optical or thermal characteristics to degrade if not detected and removed. It has also been found useful in detecting non-visible fingerprints on beryllium mirror surfaces and corrosive electrolyte on battery surface, thereby aiding in upgrading the handling and manufacturing processes” [14].

4.2 Detecting Absence/Presence of Films/Coatings

There are several applications where thin films or coatings are applied to a surface to enhance certain product characteristics or aid in further processing of parts. For example, computer hard disks and floppy disks have a very thin film of lubricant applied to enhance the life and performance of the product. In other cases, clear coatings are applied to certain surfaces to protect against corrosion or deterioration of surface finish. In many of these cases, it is important to know if the film or coating is present on the surface or not. OSEE is a quick and non-destructive method for monitoring surfaces for absence/presence of films and/or coatings.

4.2.1 Establishing Acceptance Criteria

In establishing a test method, the factors concerning the size of the OSEE sensor, static or dynamic test and contact/noncontact method need to be established first. (See [Section 4.1.1](#) for more information.) Once these factors are established, the optimum gain settings for a given test must be established. To establish the best gain setting for a given test, take OSEE readings from some coated and uncoated parts. Adjust the OSEE sensor gain setting and front panel gain setting, such that the highest possible OSEE readings are obtained from uncoated parts. The readings should not exceed 1000. Generally, uncoated parts give higher OSEE readings; however, in some cases, coated parts may give higher OSEE readings. In such cases, the gain settings should be established to give the highest readings on coated parts.

Next, a statistically large enough sample should be taken of both coated and uncoated parts. OSEE readings should be taken and recorded for each sample in both groups. Mean and standard deviation for both groups should be calculated to establish ± 3 standard deviation limits for both groups. The upper and lower

acceptance limits for coated parts should be entered into the OSEE control units. Once this is done, production parts can be tested to determine absence or presence of coating. A red light on the OSEE sensor and the front panel of OSEE control unit will indicate an uncoated part.

The purpose of taking readings on both sets of parts is to establish the usefulness of OSEE for the purpose of detecting absence/presence of coatings. In general, the further apart the readings are for the two sets of samples, the more assurance there is that the OSEE can be very effective in detecting surface films/coatings.

4.2.2 *Example Applications*

4.2.2.1 **Computer Hard Disk Manufacturing**

The computer hard disk has a very thin coating of lubricant applied as a final operation, prior to assembly in the disk drive. The presence of the lubricant film is very critical to the performance of the disk drive. The lack or absence of lubricant will result in excessive wear of read/write head, thus considerably reducing the useful life of the disk drive.

In one particular case, a customer returned over 3000 disk drives to the disk drive manufacturer because the incoming inspection discovered one disk drive where the disk was not lubricated. At this point, the manufacturer was faced with limited options. One option was to disassemble all the disk drives, strip the lubricant of all disks (since applying lubricant again without stripping the original lubricant would result in double thickness of lubricant) and reapply the lubricant. However, there was no assurance that the lubricant application would be 100% successful. The manufacturer decided to use the OSEE system to remedy the situation. The cover of each disk drive was removed and OSEE sensor was used to take OSEE readings from each disk. In a short time, the manufacturer was able to inspect all 3000 disk drive and discovered that there were three disks that were not lubricated. These three disk drives were reworked to remedy the situation. Soon after this incident, the manufacturer installed OSEE systems, in production, to monitor the absence or presence of lubricant on the disks. Several other manufacturers of hard disks have also started using the OSEE system in production for the same application.

4.2.2.2 **Aircraft Industry**

In the aircraft industry, the fuselage sheets are clear anodized to protect against corrosion. It is important to know that the sheets are anodized all over, i.e., there are no bare spots. In addition, the thickness of the anodized coating is also important. OSEE equipment is being used to check bare spots on sheets by an aircraft manufacturer. It is anticipated that after some baseline data have been collected, OSEE would be used to monitor variations in the anodized coating thickness or holes in the coating coverage.

4.2.2.3 Automotive Industry

In the automotive industry, the master brake cylinder is made out of aluminum and has a clear coating to protect against corrosion. This component is critical for safe operation of the automotive braking system. Any corrosion due to the absence of protective coating can result in brake system failure. Since the coating is not visible, OSEE is the only technique that works in a real-time environment to check the absence/presence of the protective coating to assure safe operation of the brake cylinder.

4.3 Measuring Thickness of Thin Films/Coatings

As discussed earlier, a surface without any contamination or coating usually gives a high OSEE signal, which is then attenuated by the contaminant or the coating. The attenuation of the OSEE signal is inversely proportional to the thickness of the contaminant or the coating within a certain range of the thickness of the contaminant or the coating. This range varies from material to material and the type of contaminant or coating.

The OSEE reading varies inversely with the thickness of the coating/contamination as in Eqs. (2) and (3). As the thickness increases, the OSEE reading decreases logarithmically.

Within a certain range (which varies from material to material), the thickness of a thin film or coating can be measured. A correlation must be established with some other means of measuring thickness to establish the range of OSEE readings that correspond to the desired thickness range.

The range of thickness that can be measured is dependent upon the work function of the base material, i.e., whether it is high emitting or low emitting, and the type of contamination/coating on the surface.

When the coating/contaminant is too thick, the OSEE readings generally will not change significantly as the thickness increases further. Table 3.3 is a brief list of data established on the range of thicknesses that can be measured.

TABLE 3.3 Measured Thickness Range on Certain Materials

Substrate	Coating/Contaminant ^a	Range (nm)
D6AC steel	HD-2 grease	750
Computer hard disk	Krytox or fomblin	10
Ferrous oxide coated	Lubricant	37
Silicon wafers	Silicon dioxide	80
Welding rod	Calcium content of lubricant	80

^aKrytox, manufactured by DuPont, Wilmington, DE; fomblin, manufactured by Solvay Solexis, West Deptford, NJ.

The relationship between OSEE and coatings/contamination thickness is discussed in [Section 4.3.1](#).

Typically, this range is approximately 20 nm with a maximum of 750 nm recorded on HD-2 grease on D6AC steel.

OSEE is a relative technique; hence, it gives readings in OSEE units and not in thickness units, e.g., nanometer. To determine the relationship between OSEE response versus the thickness and the range of thickness within which the relationship is proportional, tests need to be conducted on samples with known but different amounts of contaminants or coating thickness. The advantage of OSEE as a technique for measuring the thickness of thin films is that it can be implemented in-line for a real-time monitoring of thin film thickness.

4.3.1 *Establishing OSEE versus Thickness Relationship*

Again, in establishing a test method, the factors concerning the size of the OSEE sensor, static or dynamic test, and contact/noncontact method need to be established first. (See discussion in [Section 4.1.1](#) for more information.) Once these factors are established, the optimum gain settings for a given test must be established. To do this, readings should be taken from parts with no film/coating. OSEE sensor gain setting should be adjusted to obtain maximum readings but not exceeding 1000.

Next, samples should be prepared or collected to cover the range of thickness encountered during normal measurement, including samples with no films/coatings. It is important to know the film/coating thickness on each sample. These measurements should be established using some other means of measuring thickness.

OSEE measurements should be taken from each sample preferably in the same area where measurements were taken with the another method. This would prevent any errors due to variations in thickness across a sample. OSEE readings should be recorded for each sample. It is recommended that at least five samples of each thickness should be measured and an average of five samples be used in [Section 4.3.1](#).

The natural log of OSEE readings should be correlated with thickness readings obtained with the other method. Using linear regression, the best fitting straight line can be found and the slope and intercept of that line can be established. A graph should be prepared to make sure that the relationship is linear over the range of thickness measured. If the relationship is not linear over the full range, then the linear regression should be done only on the linear range. The relationship derived will be of the type:

$$Y = C - mx$$

Here, m is the slope, C is the intercept, Y is the thickness, and x is natural log of the OSEE reading. This relationship will also define the range of thickness that can be measured for this particular substrate and the type of film/coating.

Model SQM300 has the ability to measure thickness of thin films. The software has a provision to enter the slope and intercept of the correlation between OSEE and film thickness. Once this information is entered into the software, the thickness of the film on unknown samples can be measured and recorded. Since OSEE is a very fast responding technique, it is possible to install the SQM300 in-line to monitor the thickness of the film on a real-time basis, such as controlling the film thickness in the deposition process.

4.3.2 Example Applications

4.3.2.1 Computer Hard Disk

To minimize interaction of the read/write head with the magnetic disk surface, it is common for manufacturers of magnetic media to add an extremely thin layer (0.5-5 nm) from a series of commercially available fluorocarbon lubricants, to reduce stiction/friction and increase disk life durability under a wide range of conditions (typically, 300,000 contact start/stop cycles).

The application and control of a known exact thickness of fluorocarbon lubricant is of critical importance in reducing stiction/friction when considered in conjunction with two other factors, namely the type of overcoat (usually carbon) and the surface finish of the disk (average roughness) [17].

Insufficient lubricant or absence of the lubricant will result in excessive wear of read/write heads and “crashes” can occur. Too much lubricant results in “stiction,” where the read/write head sticks to the disk, possibly causing permanent damage.

From Ref [17], As an alternative to ESCA, FTIR, and ellipsometry for measuring fluorocarbon lube thickness on magnetic media, OSEE has the following advantages:

1. Results in <10 s.
2. Large throughput of disks for sampling.
3. Readings within ± 0.3 nm of FTIR.
4. Minimum operator training.
5. Compact, rugged equipment, well suited for a production environment.
6. Process variations can be identified and corrected immediately.
7. Easily used as a go/no-go test for detecting lubricant on any coated surface.

As disk drive designs call for a decrease in the flying height of the read/write heads, the lubricant overcoat becomes more and more critical; it must be precisely controlled to avoid stiction/friction and to increase durability. To attain better control, continued research must be aimed at understanding three interacting factors, namely, lubricant, texture, and magnetic overcoat. OSEE offers the capability, speed, and cost-effectiveness needed to further the understanding of these three fundamental factors [17].

4.3.2.2 Welding Wire

To maintain proper feeding of welding wire through automatic welding guns, a lubricant with calcium is applied to the welding wire. The level of lubricant thickness must be maintained within certain range. Too much lubricant results in “overfeeding” the welding wire which may result in cold/poor welds. Too little lubricant results in “underfeeding” the welding wire, resulting in voids/poor welds.

OSEE is being used to monitor lubricant thickness in-line and provide real-time control of the lubricant application process.

5 SUMMARY

Since OSEE is a fast, nondestructive and quantitative measure of the surface condition, an acceptance level of desired surface cleanliness can be established empirically or through testing to relate the OSEE readings to the surface condition. This results in downstream reduction in nonconformance and higher yield due to better measurement and control of surface cleanliness. Once the “optimum” cleanliness level has been established, a surface cleanliness measuring system can be used in process to monitor and assure that the desired cleanliness level is being achieved. By monitoring the surface cleanliness to an established level of cleanliness, the nonconformance due to surface contamination can be minimized or eliminated.

The OSEE system can be used, in-line or off-line, to monitor the effectiveness of the cleaning process. The system can also be used to optimize the cleaning process by monitoring changes in surface cleanliness as a result of changes made to the various factors that determine the effectiveness of the cleaning process, such as concentration level of surfactant, surfactant temperature, cleaning process speed, or cleaning time.

Another advantage of in-process or in-line monitoring of surface cleanliness is that replenishment of chemicals or cleaning agents will only be done when needed, and not done according to a predetermined, somewhat arbitrary schedule. This replenishment schedule is usually time dependent. In reality, the amount of contamination can vary considerably from part to part. In addition, the number of parts being cleaned during a given time frame can also vary considerably. Thus, a time-dependent replenishment schedule is not the best way of controlling the cleaning process. The required level of chemical or cleaning agent concentration in the cleaning process can also be objectively determined and maintained using surface cleanliness measuring system.

In conclusion, Photo Emission Tech’s OSEE systems are capable of determining surface cleanliness of various substrates. The technique is one of the best techniques for use in production to help answer all types of questions related to surface cleanliness.

The OSEE technique is not suitable for monitoring particulate contamination, but is well suited to thin film contamination detection/monitoring. Some of advantages of the OSEE technique for measuring surface condition are as follows:

1. It is a rapid test—it takes just a few seconds; thus, it is real-time test as opposed to sending samples to a lab and waiting for the results.
2. The test can be performed by placing the sensor in contact with the surface or in noncontact mode.
3. The system is portable and thus can be taken to the production floor for testing in-line.
4. The skill level required to operate the system is very low.
5. The system gives a visual indication of go/no-go status of any part tested.
6. Since the test is very fast, it is possible to take several measurements from each part to determine if there is any variation in the level of cleanliness for different area of the parts.
7. The technique is fast enough that, if desired, it can be installed in-line to monitor the cleanliness of parts.

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